

PROFILE

AI platform and reliability engineer with about eight years of experience delivering governed agent applications, cloud platforms, and user-facing services into production.

Builds and operates systems across agent runtime integration, delivery automation, access control, observability, incident response, and production stabilization.

SELECTED EVIDENCE

AWS EKS platform for governed, self-service AX application delivery	Production reliability for an enterprise AI service used by 30,000 employees	UK rollout, incident response, and stabilization for a global B2C generative AI service
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CORE SKILLS

Agent Platform	AWS EKS, agent runtime integration, self-service delivery, automatic domains, existing LLM Gateway API integration
Reliability	Grafana, Prometheus, OpenTelemetry, incident response, rollout stabilization, high availability, auto-scaling
Software & Delivery	Python, Bash, Linux, Kubernetes Operators, ArgoCD, GitHub Actions, GitOps, Helm/Helmfile
Security & Governance	Authentication, authorization, platform security controls, policy enforcement, enterprise governance

EXPERIENCE

Samsung Electronics, AX Development Group	Seoul, South Korea
SRE / AI Platform Engineer	Jan. 2026 - Present
- Design and build an internal AX Agent Platform on AWS EKS, adapting the agent runtime and deployment path to enterprise platform standards. - Implemented self-service AX application delivery with automatic domain issuance, authorized-only access, and platform controls for authentication, authorization, security, and governance. - Enabled AX applications to call the existing enterprise LLM Gateway API through platform-side client compatibility and runtime integration. - Supported a Microsoft-collaborative B2C generative AI service through UK rollout readiness, initial production operations, incident response, and stabilization from Jan. to Jun. 2026.	
Samsung Research	Seoul, South Korea
SRE / AI Platform Engineer	Jul. 2024 - Dec. 2025
- Led production delivery and application reliability for a company-wide AI chatbot serving 30,000 employees on internal Kubernetes clusters. - Owned releases across development, staging, and production while integrating application delivery with security and operational requirements. - Built container delivery pipelines with Docker, Kubernetes, GitHub Actions, ArgoCD, Helm, and Helmfile. - Deployed Grafana, Prometheus, and OpenTelemetry observability and built a Python/MLflow evaluation workflow for LLM-based machine translation.	
Samsung Electronics, Network Division	Seoul, South Korea
SRE & Software Engineer	Jan. 2021 - Jun. 2024
- Designed and delivered containerized applications with Docker, Helm, Kubernetes Operators, and AWS services including EKS, ECS, Route 53, RDS, and IAM. - Built and managed OpenShift, Kubernetes, and EKS environments for reliability, high availability, performance, and auto-scaling. - Designed and developed a Bash-based one-click Kubernetes cluster provisioning system for cloud developers. - Developed a Linux application that mirrors and forwards up to 130,000 packets per second under constrained system resources.	

EXPERIENCE (CONTINUED)

TmaxOS & TmaxCloud

Gyeonggi, South Korea

Software & Platform Engineer

Jan. 2019 - Dec. 2020

- Built a Siri-like AI assistant on TmaxOS, covering model inference, data preprocessing, system integration, and asynchronous application flows.
- Designed and developed Kubernetes CI/CD and registry operators and deployed a private container image registry.

SELECTED ENGINEERING WORK

Internal AX Agent Platform

Built an AWS EKS platform for governed AX application delivery with runtime adaptation, automatic domains, authorized-only access, and integration with an existing enterprise LLM Gateway API.

Observability & Release Reliability

Deployed Grafana, Prometheus, and OpenTelemetry and connected Kubernetes releases across development, staging, and production with security and operational requirements.

Enterprise AI Assistant | Public reference implementation

jinwonlee.github.io/projects/enterprise-ai-assistant/web/

Built an access-aware Python assistant with SQLite FTS5 retrieval, pre-generation role filtering, grounded citations, audit events, evaluation cases, a FastAPI surface, and an optional OpenAI Responses API provider.

Global B2C Generative AI Service

Supported UK rollout readiness, initial production operations, incident response, and early stabilization for a Microsoft-collaborative consumer AI service on Azure.

EDUCATION & RESEARCH

UNIST | M.S. in Computer Science and Engineering

Mar. 2017 - Feb. 2019

Master's thesis: Convergence Aware CNN Training | Advisors: Jiwon Seo and Sam H. Noh

UNIST | B.S. in Computer Science and Engineering, Minor in Mathematics

Mar. 2012 - Feb. 2017

Undergraduate research: distributed systems for AI model training with TensorFlow

PUBLICATIONS & LANGUAGE

- USENIX NSDI 2019 poster: Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement.
- arXiv 2019: Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement.
- ACM EuroSys 2018 poster: Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism.
- OPic IH (Intermediate High), Dec. 2025.